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320733(20)

B. E. (Seventh Semester) Examination, April-May 2019

(New Scheme)

(Civil Engg. Branch)

ENVIRONMENTAL ENGINEERING–II

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Answer all **five** questions. Part (a) of each question is compulsory and carries 02 marks. Attempt any **two** parts from (b), (c) & (d) and carries 07 marks.*

Unit-I

1. (a) Manholes are generally located : 2

(i) at all changes of direction of sewer

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(ii) at all changes of gradient of sewer

(iii) at al Junctions of different sewers

(iv) all of the above

(b) Describe the various system of sanitation giving their advantages considering the water shortage in townships and citieis, what are the modifications you would suggest for better functioning of system. 7

(c) Design a sewer to serve a population of 36,000, the daily per capita water supply allowance being 135 litres, of which 80 percent finds it's way in to sewer. The slope available for the sewer to be laid is 1 in 625 and the sewer should be designed to carry four times the dry weather flow when running full. What would be the velocity of flow in the sewer when running full?

(assume $n = 0{\cdot}012$ in Manning's formula) 7

(d) Derive the expression for shields self cleansing velocity. Also derive expression for proportionate area, velocity and HMD. 7

Unit-II

2. (a) The settling velocity of a particle in a sedimentation tank depends on : 2

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(i) depth of tank

(ii) surface area of tank

(iii) both depth and surface area of tank

(iv) none of these

(b) The 5 day 30°C BOD of sewage sample is 110 mg./l. Calculate its 5 days 20°C BOD. Assume the Deoxygenation constant at 20°C, K 20 as 0·1. 7

(c) Design the dimensions of a septic tank for a small colony of 150 persons provided with an assured water supply from the municipal head works at a rate of 120 litres per person. Assume any data, you may need. 7

(d) Explain Purifying action of an oxidation pond. 7

Unit-III

3. (a) Define attached growth process and suspended growth process. 2

(b) Explain the activated sludge process with a schematic diagram. 7

(c) Define with respect to biological waste water treatment : 7

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(i) MLSS

(ii) MLVSS

(iii) F/M Ratio

(iv) SVI

(d) Determine the size of a high rate trickling filter for the following :

Flow	=	4·5 Mld
Recirculation ratio	=	1·4
BOD of raw sewage	=	250 mg/l
BOD removal in primary clarifier	=	25%
Final effluent BOD Desired	=	50 mg/l

Calculate also the size of standard rate trickling filter to accomplish the above requirement. 7

5.

Unit-IV

4. (a) Define stream standards and effluent standards. 2

(b) Explain four zones of pollution in the river and how self purification is achieved by the river. Explain the factors affecting self purification. 7

(c) Describe the characteristics of waste generated from

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a sugar factory and also mention the treatment methodology to be adopted for such waste. 7

(d) Write short nortes on : 7

(i) Oxygen sag curve

(ii) Dilution method Vs. Land disposal method for sewage disposal

Unit-V

5. (a) Differentiate "Garbage and Refuse". 2

(b) Explain 4 RS of solid wate management, in detail. 7

(c) Explain the social and helath Implications of improper solid waste management. 7

(d) Describe the mechanical composting techniques adopted in India :

(i) Banglore method of composting

(ii) Indore method of composting

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